Bharat Singh Bhati

Software Developer | Machine Vision | IoT | On-Device AI

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Professional Summary

Computer Science Engineering graduate with hands-on experience building AI, Machine Vision and IoT systems. Experienced in Python, C++, PyTorch, OpenCV, Whisper, YOLO, embedded systems, Linux and Docker. Built fully offline AI assistants, embedded IoT solutions and computer vision applications with a strong focus on edge computing and real-world deployment.

Technical Skills

Languages: Python, C++, Java, SQL

AI & Machine Learning: PyTorch, OpenCV, YOLO, Faster-Whisper, NumPy, Pandas, Ollama

Embedded & IoT: ESP32, Arduino, Raspberry Pi, MQTT, Sensor Fusion, Edge Computing

Systems: Linux (Arch, Debian), Docker, Git, Bash, VS Code

Concepts: REST APIs, Flask, OOP, DSA, DBMS

Experience

IoT Developer – RAIOT Labs, Amity University Jaipur 2022–2026

- Built five Python-based IoT prototypes including a robotic arm and air-quality monitoring system deployed on Raspberry Pi.
- Mentored more than twelve junior developers in Python, embedded systems and sensor programming.
- Presented four working prototypes during university technical exhibitions.

Projects

CyberWolf AI Assistant

- Developed an offline AI assistant using PyTorch, Faster-Whisper, Llama 3.2 and OpenCV.
- Trained a custom wake-word model achieving over 95% accuracy in quiet environments.
- Implemented local speech recognition, LLM inference and face recognition for complete offline authentication.

LumiSense

- Designed an IoT-based smart ambient lighting system using Arduino and C++.
- Integrated gas, CO and particulate sensors with MQTT communication.
- Implemented real-time monitoring with automated alerts.

Network Security Auditor Tool

- Developed a Python-based network scanner using Netdiscover and socket programming.
- Automated topology mapping and MAC filtering analysis across lab environments.

EDUCATION

B.Tech. Computer Science and Engineering

Amity University Rajasthan, Jaipur

CGPA: 7.9/10 2022–2026

ACHIEVEMENTS

- 1st Place – IEEE Hackathon 2024
- 2nd Place – IEEE Hackathon 2023
- Lead Organizer – Sun Bulls Future Energies Pvt. Ltd Event
- Participant – IIT Roorkee Cognizance 2025

CERTIFICATIONS

- Prompt Engineering for Developers – DeepLearning.AI (Andrew Ng)
- Robotron Robotics Course – Tech Analogy (DPIIT Recognised)